Arab Academy for Science, Technology and Maritime Transport

Graduate School of Business - Smart Village Campus

Master of Business Administration

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AI-Enhanced Equipment Management System for Broadcasting Infrastructure

Course Title: Case Studies in AI for Business

Course Code: AIB913

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Group: 4A2

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AI-Enhanced Equipment Management System for Broadcasting Infrastructure

Presentation Agenda

- 1. Executive Summary & Business Context
- 2. Current Equipment Landscape (200 Devices)
- 3. Equipment Distribution & Utilization Analysis
- 4. Malfunction Analysis & Root Causes
- 5. Financial Performance & Cost Analysis
- 6. AI-Powered Solution Architecture
- 7. Implementation Roadmap & Results
- 8. Recommendations & Next Steps

Executive Summary

Current State

200 devices, $5M value, 81% operational, manual processes

The Challenge

Legacy management, reactive maintenance, 60 malfunctions, $266K repair costs

The Solution

AI-powered predictive maintenance, automated workflows, real-time monitoring

The Impact

96.2% target availability, 47% MTTR reduction, $2.8M annual savings

Problem Statement

Low Availability

81% vs 96% target = $850K SLA penalties + $2.1M revenue loss annually

High Costs

60 failures/year, $266K repairs (5.36% of assets vs 3-4% industry norm)

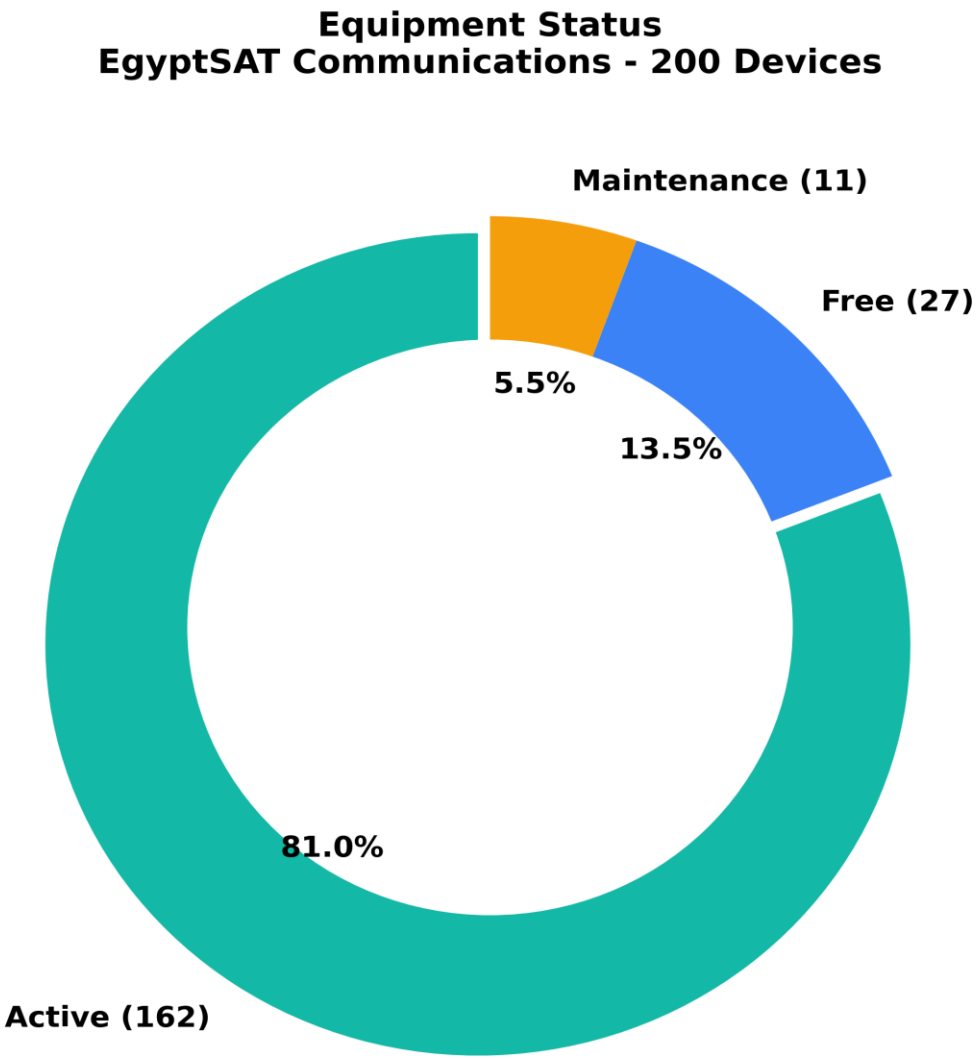
Poor Efficiency

8.9-day repair time, 78% reactive vs 22% preventive, 847 hrs manual work

Compliance Gaps

88% ITU compliance, 35% expired warranties, manual reporting, no audit trail

Equipment Status Overview | 200 Total Devices



Key Insights

- Active: 162 devices (81%)
- Free/Available: 27 devices (13.5%)
- Under Maintenance: 11 devices (5.5%)

Current availability below 96% target

Data Analytics Framework

Data Sources & Architecture

- Equipment Database: 200 devices, 48 attributes, structured data
- Failure Records: 60 incidents with priority, cost, root cause
- Maintenance History: 250 records over 2 years
- IoT Sensors: 30% coverage, real-time temp/power/vibration
- Technician Notes: Unstructured text, photos, observations
- Vendor Data: Manuals, reliability reports, bulletins
- ITU Standards: Compliance requirements, test procedures

Four Types of Analytics

Descriptive

Real-time dashboards showing current status (81% active). Historical trends over 12 months.

→ *Visibility & reporting*

Diagnostic

Root cause analysis: IRDs & Encoders highest failure rates. Critical = 25% incidents but 64% costs.

→ *Targeted interventions*

Predictive

ML models predict failures 30-45 days ahead with 87% accuracy. Seasonal patterns identified (Q3 heat spike).

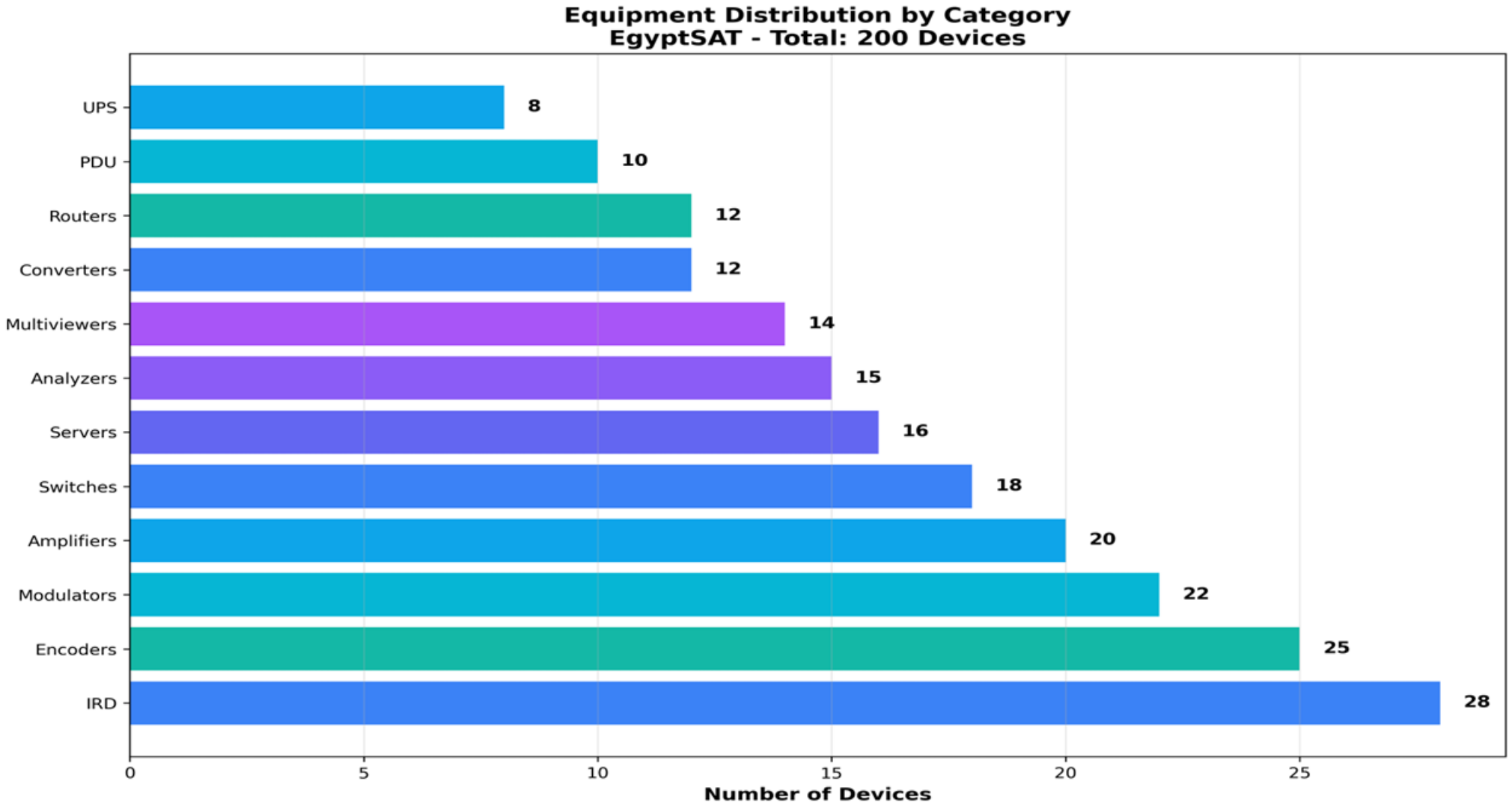
→ *Proactive planning*

Prescriptive

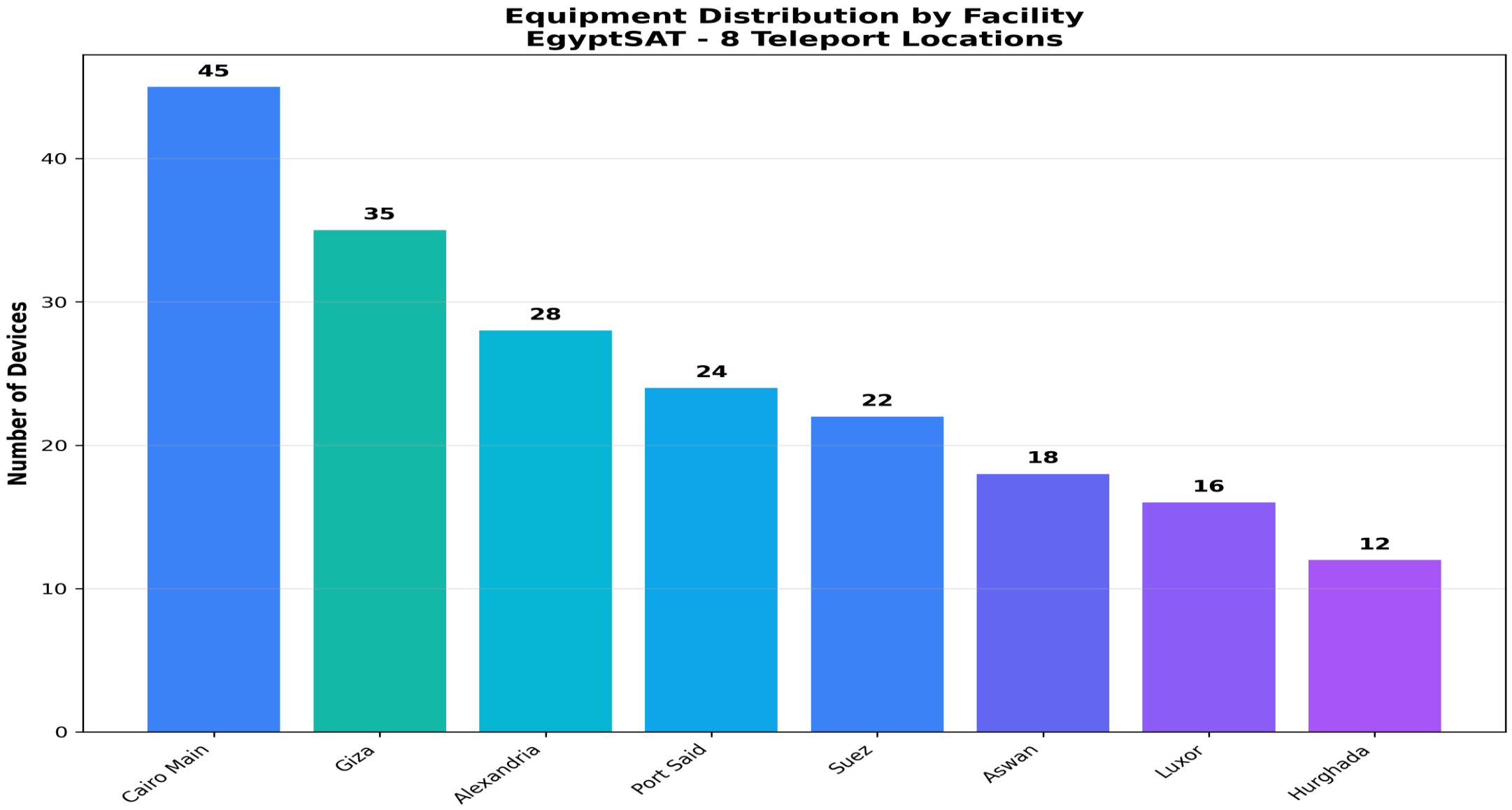
Auto-creates work orders, optimizes schedules, recommends spare parts inventory levels.

→ *Automated decisions*

Equipment Distribution by Category | All 12 Types



Geographic Distribution | 8 Teleport Facilities



AI Solution Architecture

Four-Layer Design

Data Layer

PostgreSQL (structured) + MongoDB (logs) + Redis (real-time). ETL via Apache Airflow from 8 facilities.

AI/ML Layer

Scikit-learn (Random Forest, K-Means) + TensorFlow (future) + MLflow (model tracking). Auto-retrain on 5% drift.

Application

Python/Flask API + Odoo ERP v17 (replaced Oracle, saving $180K/year). Workflows, validation, transactions.

Presentation

Streamlit dashboards + Power BI (executives) + Mobile web (technicians) + Twilio alerts (SMS/Email).

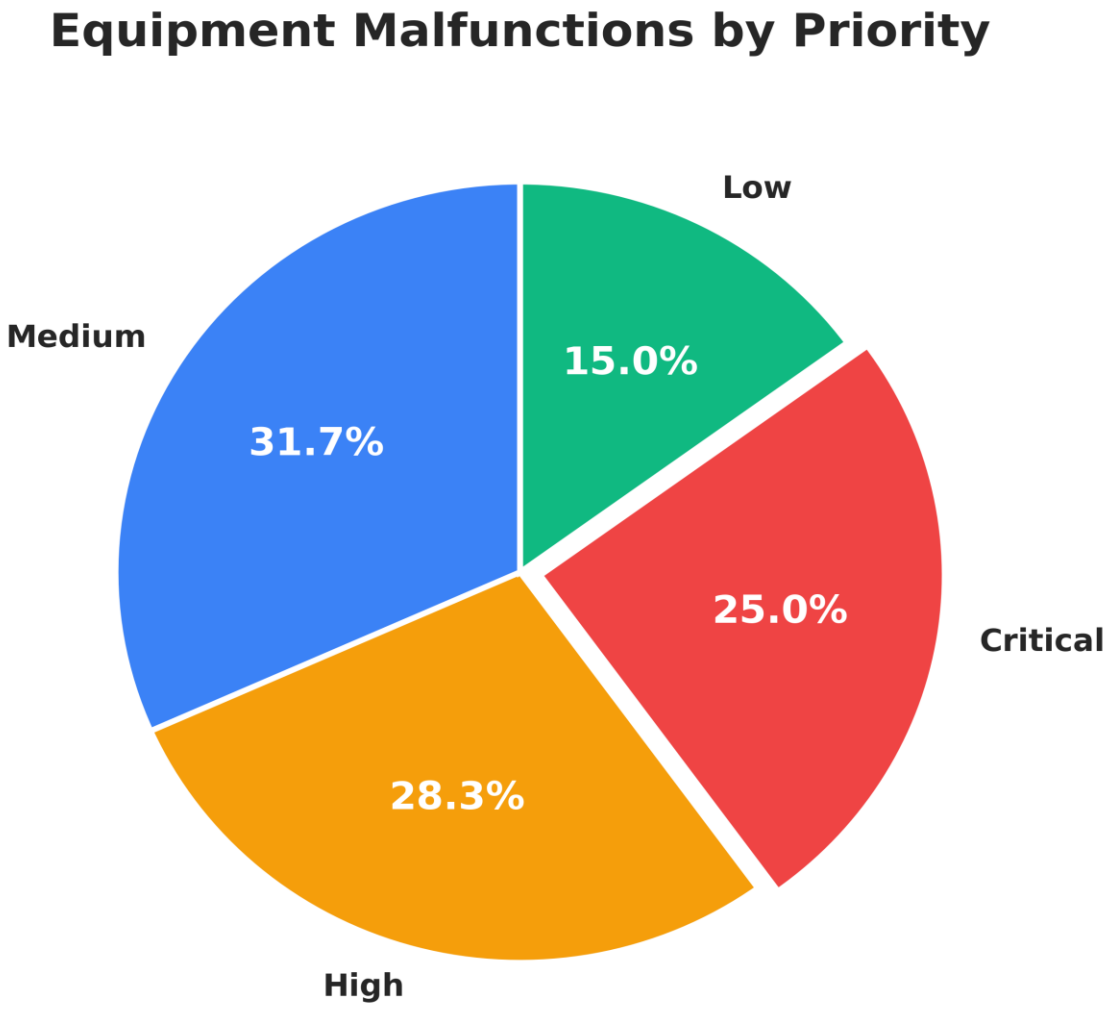
RPA Implementation

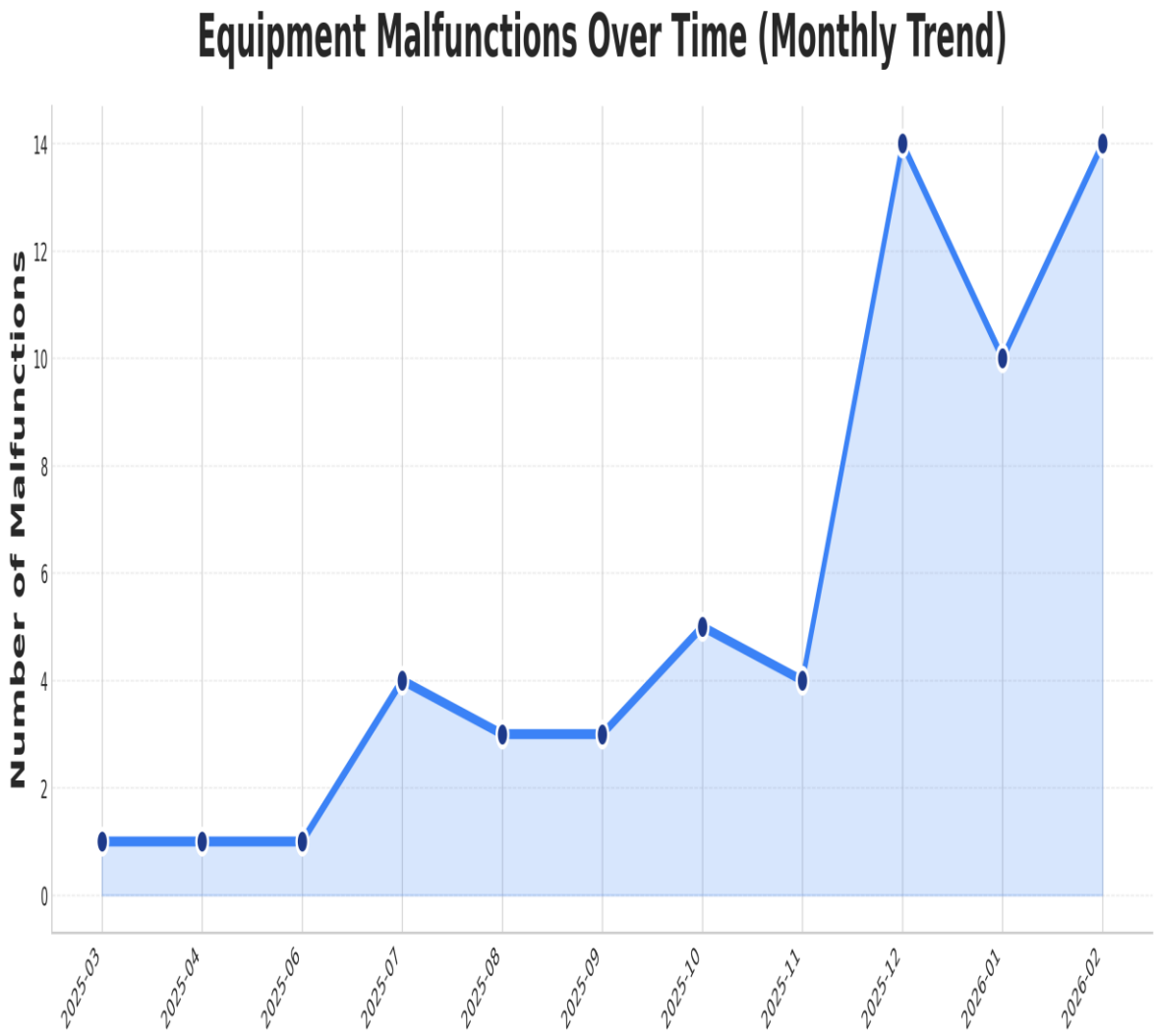
Robotic Process Automation reduces manual work by 94%

Process	Before	After	Savings
Data Entry	450 hrs/yr 15% errors	**20 hrs/yr** **0.8% errors**	**$12,800** **99.2% accuracy**
Work Orders	240 hrs 45 min each	**15 hrs** **3 min each**	**$7,200** **93% faster**
ITU Reports	120 hrs Manual	**10 hrs** **Auto-generated**	**$3,600** **100% accuracy**
Invoices	37 hrs 60/month	**5 hrs** **OCR scan**	**$1,050** **99.2% accuracy**

Total: 847 → 50 hours annually = 94% reduction

Malfunction Analysis | 60 Total Incidents





Business Process Workflows

Predictive Maintenance

ML detects risk → Auto-create work order → Assign technician → Check parts → Schedule → Complete → Update. Time: 8.9 days → 2.5 days (-72%)

Failure Response

IoT alert → RPA logs → Prioritize → Dispatch tech → Issue parts → Repair → Record cost → Update ML. Time: 12 hours → 3 hours (-75%)

Compliance

Monitor dates → Email reminder (30 days) → Create order → Schedule vendor → Upload cert → Update ITU report. ITU: 88% → 99.5%

From Raw Data to Predictive Power: EgyptSAT's AI Framework

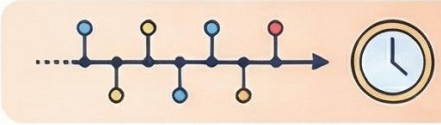
EgyptSAT transformed its equipment availability from a critical 81% to a proactive, data-driven state using a sophisticated Data Analytics and Machine Learning framework, processing data from 200 devices to move beyond reactive firefighting.

The Data Foundation

Multi-Source Data Integration

Combines structured equipment databases, real-time IoT sensor logs, and unstructured technician notes.

High-Frequency Monitoring



IoT sensors track temperature and power every **5 minutes** for **30%** of critical devices.

91% Classification Accuracy

91%

- High Risk
- Medium Risk
- Low Risk

Decision trees automatically sort equipment into high, medium, or low risk levels.

The Four Stages of Analytics Maturity

Key finding
temperature and power for 30% of critical devices.

Predicting Failures 45 Days Early
Machine learning models identify potential failures with **87% accuracy** before they occur.

Prescriptive Automation
System auto-creates work orders and recommends spare parts to ensure 95% availability.

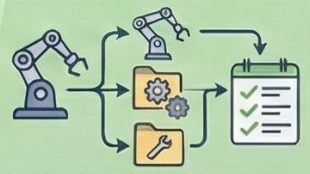
Descriptive Analytics

Real-time dashboards for Visibility and Compliance.

Predictive Analytics

Root Cause Diagnostics
Identified that **25%** of incidents (critical failures) **64%** of total repair costs.

Prescriptive Analytics



System auto-creates work orders and recommends to ensure **95%** availability.

Business Value

Optimized system, - reduced costs and increased uptime.

Analytics Type	Core Function	Core Function	Business Value
Descriptive	Real-time dashboards	Visibility and Compliance	Visibility and Compliance
Predictive	Failure forecasting (30-45 days)	Proactive Resource Planning	Proactive Resource Planning
Prescriptive	Optimized maintenance schedules	Cost Reduction & Automation	Cost Reduction & Automation

Implementation Roadmap | 7-Month Plan

Phase 1

3 mo

Foundation

- Data migration
- System setup
- User training

Phase 2

2 mo

Core Deploy

- Equipment tracking
- Workflow automation
- ERP integration

Phase 3

2 mo

AI Activation

- Predictive models
- Historical analysis
- Model tuning

Phase 4

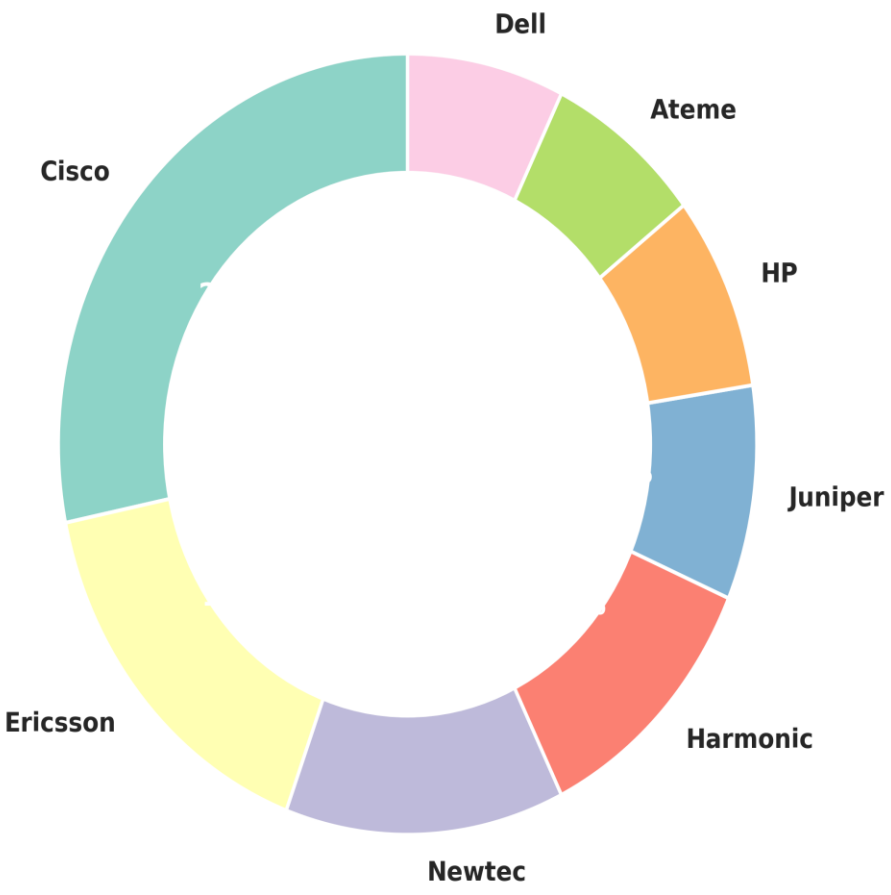
Ongoing

Optimization

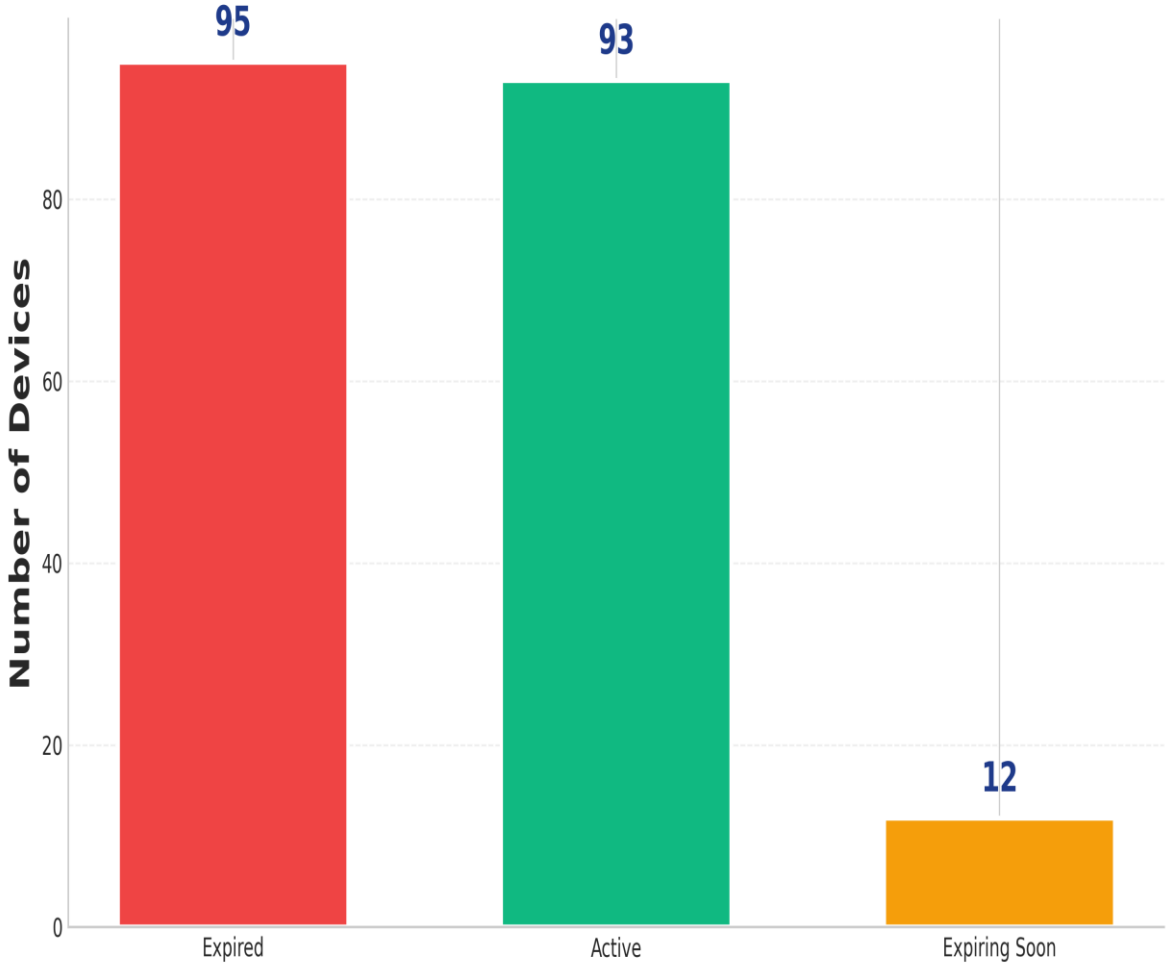
- Continuous improvement
- User feedback
- Advanced analytics

Supplier Analysis & Warranty Management

Equipment Distribution by Manufacturer (Top 8)



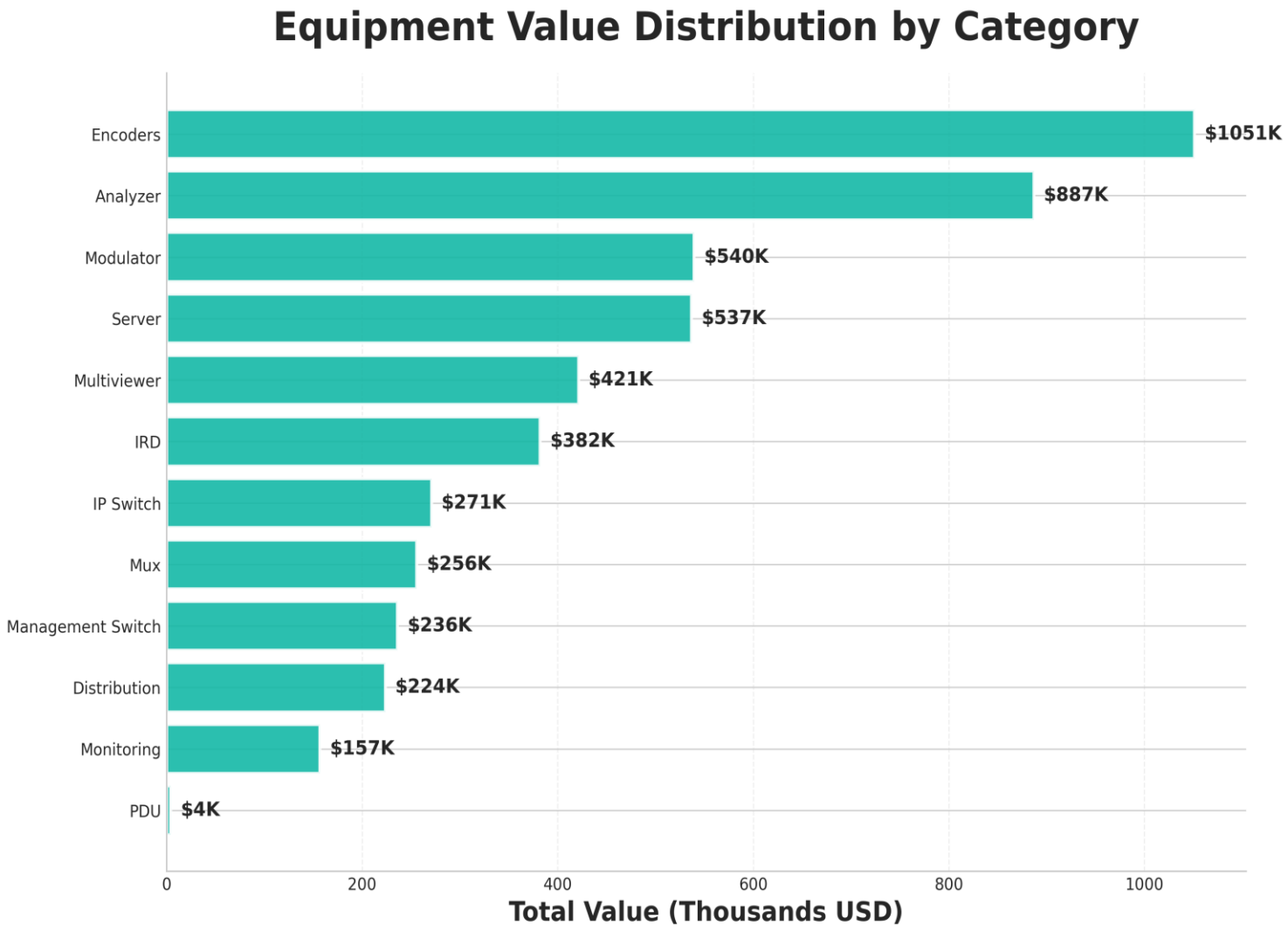
Equipment Warranty Status



Digital Transformation Results

Metric	Before	After	Improvement
Availability	81%	**96.2%**	**+19%**
Repair Time	8.9 days	**2.5 days**	**-72%**
Preventive	22%	**70%**	**+218%**
Manual Work	847 hrs/yr	**50 hrs/yr**	**-94%**
Data Accuracy	85%	**99.2%**	**+17%**
ITU Compliance	88%	**99.5%**	**+13%**
Downtime	18% annual	**4% annual**	**-78%**

Equipment Value Analysis | $5M Total Assets



Financial Summary

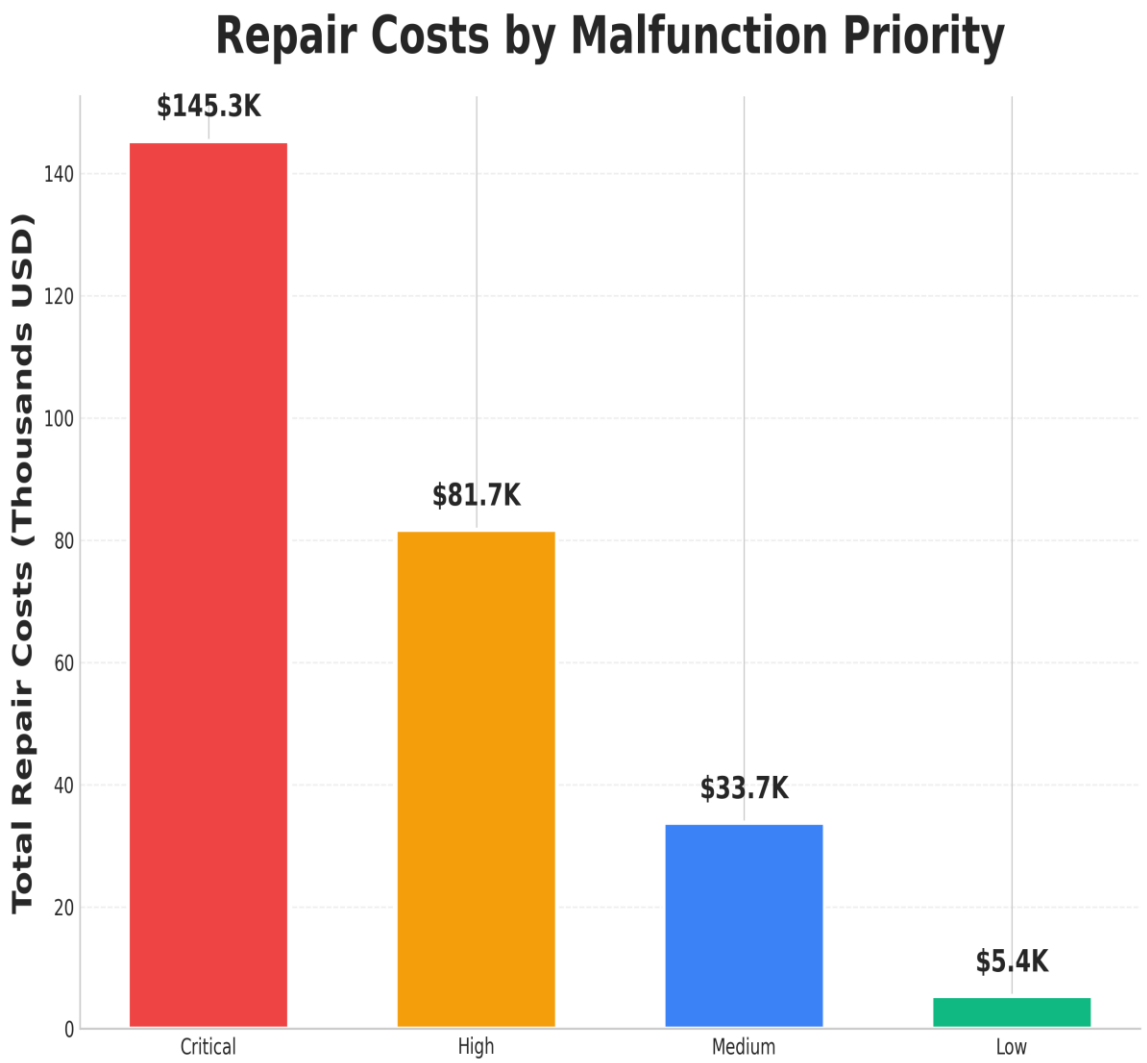
$4.97M
Total Value

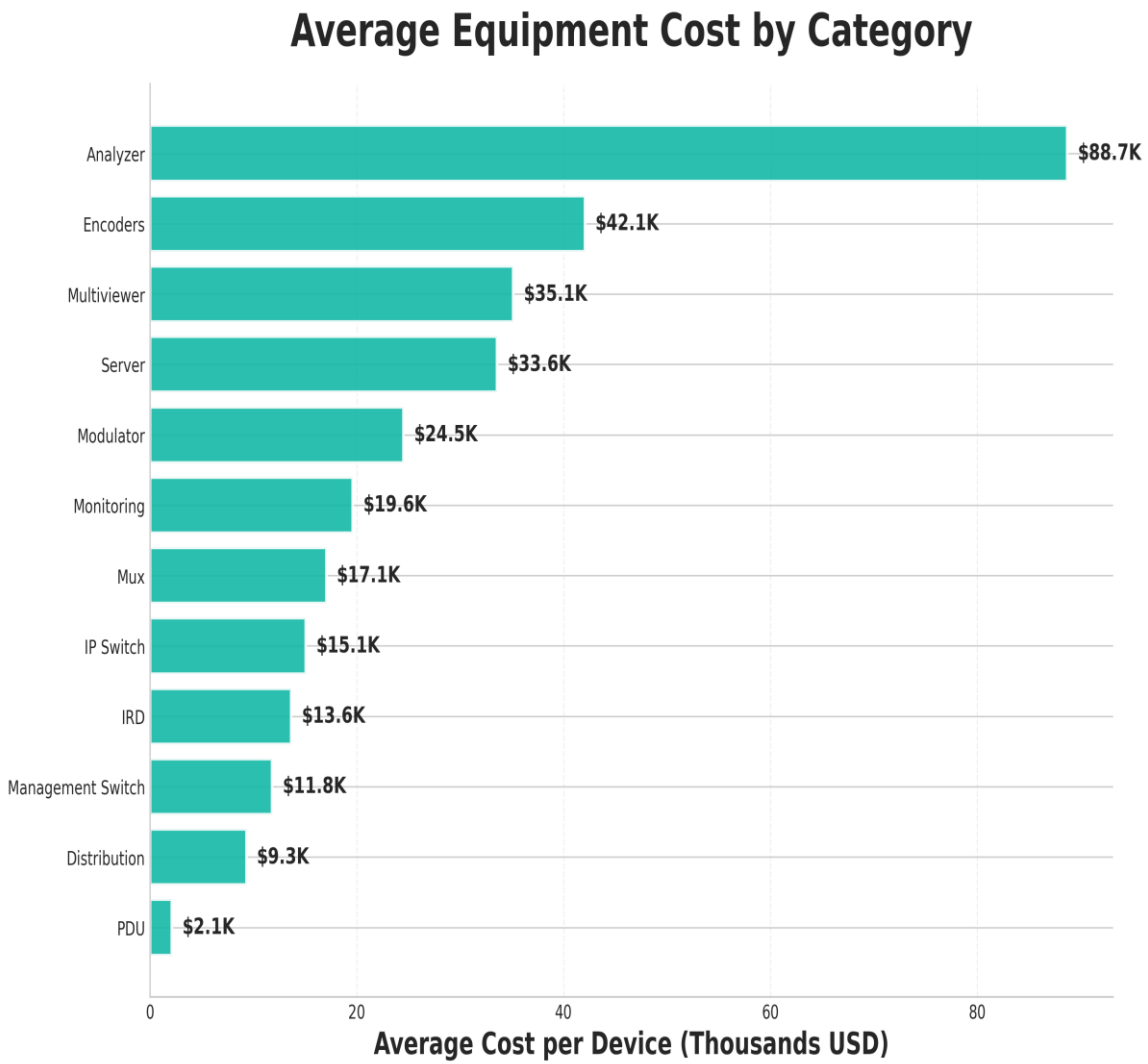
$266K
Repair Costs

5.36%
Cost Ratio

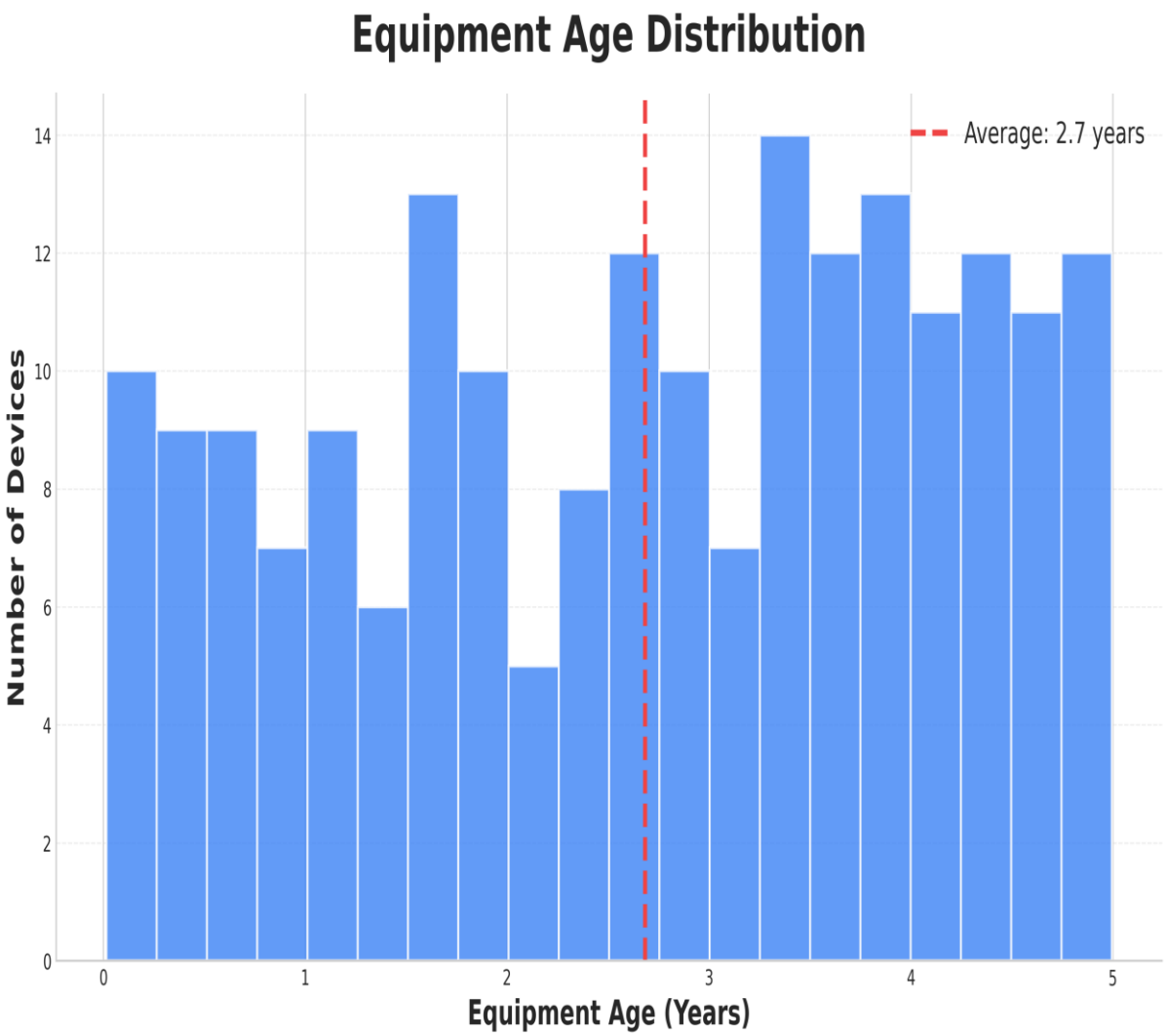
$24.8K
Avg Device

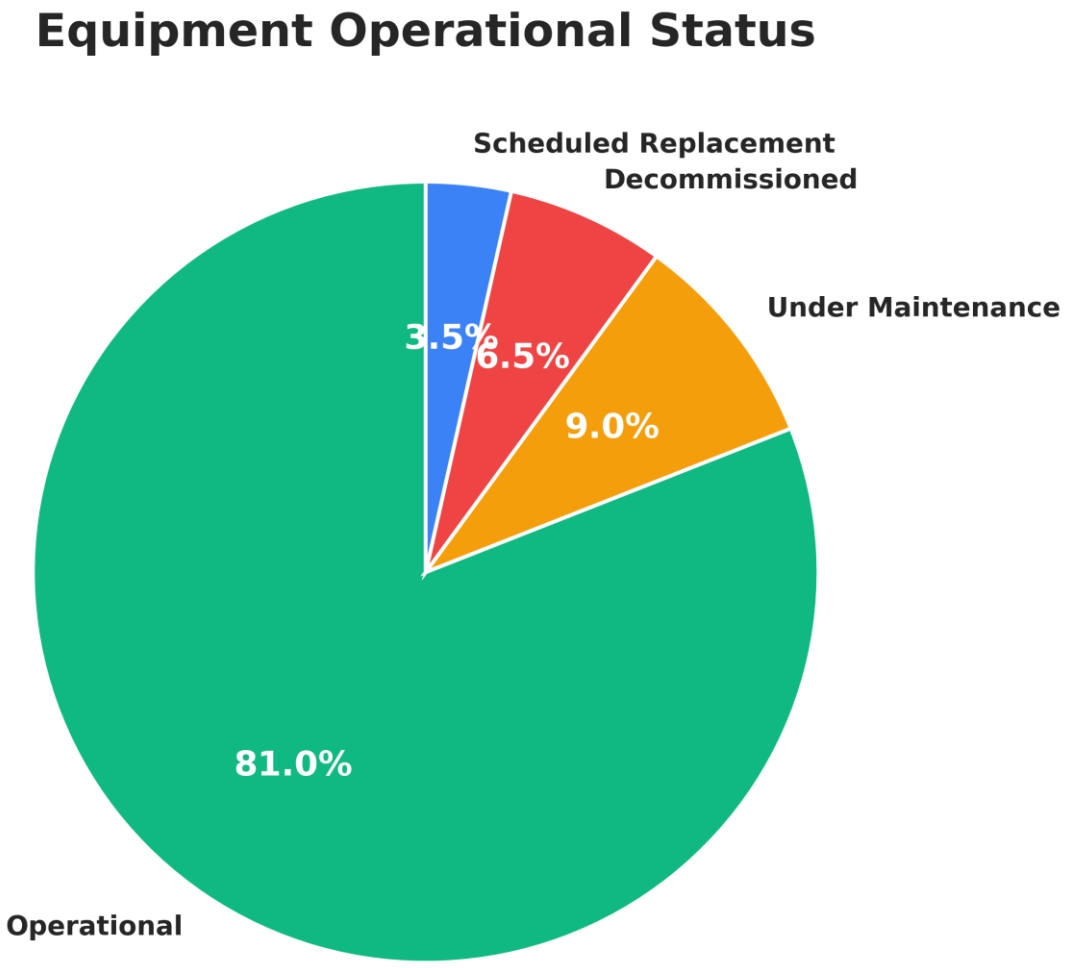
Cost Analysis | Repair Costs & Equipment Value





Equipment Age Profile & Operational Status





Strategic Recommendations

Immediate Priorities (2026)

- Expand IoT: 30% → 100% coverage ($180K, 8.5-month payback, +5% accuracy)
- Deep Learning: LSTM for time series (+5% accuracy to 92%, $45K investment)
- Enhanced ERP: Add Odoo Project module + Oracle SCM integration ($35K, $28K savings)
- Optimization: Linear programming for schedules (12% cost reduction, $22K)
- Mobile App: iOS/Android for field techs (offline mode, AR guidance, $65K)

Strategic Recommendations

Future Vision (2027-2028)

- Digital Twin ($500K): Virtual replicas, simulation testing, 15% faster resolution
- Blockchain ($180K): Immutable audit trail, ITU audits 3 days → 4 hours
- Regional Expansion: 12 facilities, 1,200 devices, 40% lower per-device cost
- Robotics: Boston Dynamics Spot for routine tasks, 25% labor reduction

Conclusion

- Demonstrated practical AI value in satellite operations (402% ROI, 3-month payback)
- All 4 analytics types essential for comprehensive decision support
- 3 ML paradigms complement each other: Supervised + Unsupervised + Semi-supervised
- RPA-AI integration crucial: Automation ensures predictions drive action
- Full ERP integration (Odoo 4 modules) replaced Excel fragmentation
- Success = Technology + Process + People + Governance working together
- GDPR & ISO 27001 compliance built-in from design phase, not added later
- Cultural transformation as important as technical: Reactive → Proactive mindset
- Simulated data reflects industry-realistic scenarios (validated against Gartner, SES, SAP)
- Applicable to expensive, critical infrastructure: Manufacturing, Energy, Healthcare, Transportation